

TED (15) – 3261

Reg. No.

(Revision - 2015)

Signature

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/

TECHNOLOGY (Model Question Paper)

Foundations of Computer Software

(Information Technology)

[Time: 3 hours]

(Maximum marks: 100)

PART – A

Marks

- I. Answer the following question in one or two sentences. Each question carries 2 marks
1. Define Abstract Data Type
 2. Define DE-Queue
 3. Name two graph traversal methods.
 4. Define process control block
 5. List any two page replacement algorithms (5x2=10)

PART – B

- II. Answer *any five* of the following. Each question carries 6 marks.
1. Explain the concept of two dimensional arrays.
 2. Write the algorithm to search an element in a singly linked list
 3. Write a note on (1). Leaf node (2). Degree of a node in Tree
 4. Write the algorithm of linear search
 5. Compare application and system software
 6. Explain time sharing system
 7. Explain virtual memory (5x6=30)

PART – C

(Answer *one* full question from each unit. Each question carries 15 marks.)

Unit – I

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|------|--|---|
| III. | a. Define data structure. Explain about any two basic data structures. | 7 |
| | b. Write the algorithm to insert an element in a linked list. | 8 |

OR

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|-----|---|---|
| IV. | a. Explain stack and its various operations. | 7 |
| | b. Write the algorithm to insert in circular queue. | 8 |

Unit – II

- V. a. What is binary search? Explain with an example. 7
- b. Write the algorithm for post-order traversal of a binary tree. 8

OR

- VI. Write the algorithm of quick sort and explain. 15

Unit - III

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|------|---|---|
| VII. | a. Explain about multi-programmed system. | 7 |
| | b. Explain about priority scheduling algorithm. | 8 |

OR

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|-------|---|---|
| VIII. | a. Explain batch processing system. | 7 |
| | b. Explain about Round Robin algorithm. | 8 |

Unit – IV

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|-----|---|---|
| IX. | a. Explain Least Recently Used (LRU) algorithm. | 8 |
| | b. Explain about tree structured directory. | 7 |

OR

- X. Differentiate various directory structures. 15